

Dart Fundamentals

Dart Control Flow & Functions

Dart: Control Flow & Functions

Dart Fundamental



By the end of this lecture, you should be able to

- Understand Conditional Statements.
- Loops.
- Functions (Methods).
- Practical Examples & Exercises.

Conditional Statements

Used to make **decisions** in code.

1. **Conditional Statement** → Allows execution of different code blocks based on whether a condition evaluates to true or false (e.g., if, else if, else).
2. **Switch Statement** → Evaluates an expression and executes code blocks based on matching cases, typically used with values of the same type (e.g., int, String, or enums in Dart).
3. **Loops** → Repeatedly execute a block of code as long as a condition is met (e.g. for while, do-while)

```
void main() {  
    int age = 18;  
  
    if (age < 18) {  
        print("You are underage");  
    } else if (age == 18) {  
        print("You just became an adult");  
    } else {  
        print("You are an adult");  
    }  
}
```

Loops

A for loop is used to repeatedly execute a block of code a specific number of times, typically when you know the number of iterations in advance. It consists of three parts: initialization, condition, and update.

```
for (initialization; condition; update)
{
    // Code to execute
}
```

```
for (int i = 1; i <= 5; i++) {
    print("Number $i");
}
```

While Loop

A **while loop** repeatedly executes a block of code as long as a condition is true. It's useful when the number of iterations is unknown.

```
while (condition) {
    // Code to execute
}
```

```
int i = 1;
while (i <= 5) {
    print(i);
    i++;
}
```

Functions (Methods)

What is a Function?

A **function** is a block of code that performs a specific task.

Instead of repeating code, you define it **once** and call it **anytime you need** it.

Functions help you:

- Write **organized**, **reusable**, and **clean** code.
- Break large programs into smaller, manageable pieces.

Basic Function

Mini Code Tryout

Examples for the day

1. Check if a number is positive, negative, or zero.
2. Print all even numbers from 1–20 using a loop.
3. Write a function that returns the square of a number.
4. Create a function that takes two names and returns a greeting like
`"Hello, Alice and Bob!"`
5. Use a `switch` statement to print the day of the week based on a number (1–7).

Home Practice

Combine what they've learned by building a **small calculator** using:

1. `if` or `switch` for operation selection
2. functions for each operation
3. loops to allow continuous use until "exit" is typed